Simulation-Based Inference For An Adaptive-Network Epidemic Model

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§1 Introduction

Accurate modeling of infectious disease spread remains a central problem in epidemiology. Classical SIR frameworks assume homogeneous mixing or static contact networks (Anderson & May, 1991), yet individuals routinely adapt their behaviour during an epidemic — avoiding infected contacts or seeking new ones — in ways that directly alter transmission pathways (Gross et al., 2006). The adaptive-network SIR model captures this feedback by coupling epidemic dynamics with network evolution: susceptible nodes may rewire connections away from infected neighbours, reflecting behavioural avoidance. This co-evolution of states and structure creates a nontrivial identifiability problem, as distinct combinations of infection, recovery, and rewiring rates can produce similar observed dynamics.

We consider the problem of inferring model parameters from partially observed epidemic realisations. The likelihood of the observed data is intractable, requiring integration over all latent network configurations and epidemic trajectories under path-dependent dynamics. We therefore employ Approximate Bayesian Computation (ABC) (Beaumont et al., 2002), which replaces likelihood evaluation with simulation-based comparison of summary statistics.

This report proceeds as follows. We begin with rejection ABC as a baseline posterior approximation, then examine summary statistic construction under the semi-automatic framework of (Fearnhead & Prangle, 2012) and evaluate performance across methods. Finally, we apply neural posterior estimation (NPE) as a more sample-efficient alternative (cite).

§1.1 Model

We consider an adaptive-network SIR model on a contact network of $N = 200$ nodes, with edges representing potential transmission pathways. At each discrete time step, three processes occur in parallel: susceptible nodes (S) adjacent to infected nodes (I) become infected with probability β; infected nodes recover with probability γ; and each S–I edge is rewired with probability ρ, redirecting the susceptible node toward a uniformly chosen non-neighbour. Recovered individuals (R) remain in the recovered state. The three parameters and their priors are summarised in Table 1.

Parameter	Description	Prior
β (infection)	Each susceptible neighbor of an infected agent becomes infected with probability β. New infections are applied after all attempts are evaluated.	Unif(0.05, 0.50)
γ (recovery)	Each infected individual (excluding those infected in the current time step) recovers with probability γ.	Unif(0.02, 0.20)
ρ (rewiring)	For each S-I edge, with probability ρ, the connection is rewired to a uniformly random non-neighbour.	Unif(0.0, 0.80)

Table 1: Model parameters and network evolution dynamics.

§1.2 Simulation

The network is initialised as an Erdős–Rényi graph $G(200, 0.05)$ with $n_0 = 5$ infected nodes and simulated for $T = 200$ time steps. Observational data are generated from $R = 40$ independent replicates under the same unknown parameter vector (β, γ, ρ). For each replicate, the following are recorded: the time series of the infected fraction $I(t)/N$, the rewiring event count at each step, and the final degree distribution of the network. Summary statistics for ABC are derived from these quantities; the goal is to recover the true (β, γ, ρ) governing the epidemic.

To improve the computational efficiency of the adaptive network model simulator, we reimplemented it in `C++` and ran the simulations in parallel on a local machine, which allowed us to generate the large number of simulation ($N = 10^5$) required for ABC inference in a reasonable timeframe ($\sim$ 25 minutes). These simulations were stored in a reference table for subsequent ABC analyses, which allowed us to experiment with different summary statistic designs and distance metrics without needing to rerun the simulations. The code for the C++ implementation and parallel simulation is provided in Appendix A.

§2 Basic Rejection ABC

We begin with basic rejection ABC with few summary statistics as a baseline for inference on (β, γ, ρ).

Algorithm 1

1. Calculate summary statistics $s = S(D)$ from observed data
2. For $t = 1$ to $t = N$,
   i. Sample θ'_t from the prior distribution $p(\theta)$
   ii. Simulate data D_t'' from the model, $p(D|\theta'_t)$
   iii. Calculate summary statistics $s'_t = S(D'_t)$ from simulated data.
3. Scale s and s'_t for $t = 1$ to $t = N$ appropriately (e.g. mean zero and unit variance).
4. For $t = 1$ to $t = N$, calculate distance and accept $d_t = d(s, s'_t) \leq \delta_\varepsilon$, with the scaled summaries.
5. Estimate posterior density $p(\theta|s)$ from the εN accepted parameters θ'_t.

As $\varepsilon \to 0$, the approximate posterior density $p(\theta|s)$ converges to the true posterior $p(\theta|D)$, but the acceptance rate becomes very low, leading to high computational cost. In practice, we choose a small but non-zero ε to balance accuracy and efficiency.

§2.1 Setup

We performed $N = 10^5$ simulations, drawing parameters independently from the prior at each iteration. For each draw, $R = 40$ independent replicates were simulated and a vector of summary statistics was computed and averaged across replicates. The statistics capture epidemic dynamics — area under the infected fraction curve, peak infected fraction, and time to peak infection — as well as network restructuring: total and peak rewiring counts, time to peak rewiring, and the mean and standard deviation of the final degree distribution. The choice of these summary statistics is somewhat arbitrary, but they were chosen to capture key features that are likely to be informative about the parameters. Acceptance was based on the Euclidean distance between observed and simulated summary vectors, with acceptance rate $\varepsilon \in \{0.1, 0.01, 0.001\}$. Full descriptions and motivation for the summary statistics are provided in Appendix A (see Table 7).

§2.2 Results

Overall, the rejection ABC results indicate that the parameters are identifiable from the chosen summary statistics, as shown by the strong contraction of the posteriors relative to the priors across all ε thresholds and the stability of posterior summaries under decreasing tolerance (see Figure 1). The joint posterior reveals strong positive dependence between β and ρ, see Figure 2, characterised by a narrow ridge of high posterior density, while γ appears more weakly constrainted. This suggests partial non-identifiability due to structural confounding, since both parameters similarly affect early epidemic growth and peak dynamics and can therefore compensate for one another.

ε	Parameter	Mode	Mean	Median	95% HDP
0.1	β	0.1699	0.2181	0.2055	(0.0753, 0.3849)
	γ	0.1020	0.1261	0.1254	(0.0606, 0.2000)
	ρ	0.2826	0.2578	0.2635	(0.0368, 0.4637)
0.01	β	0.1563	0.1535	0.1529	(0.0981, 0.2129)
	γ	0.0990	0.1174	0.1126	(0.0615, 0.1804)
	ρ	0.2689	0.2545	0.2573	(0.1273, 0.3857)
0.001	β	0.1304	0.1376	0.1347	(0.1066, 0.1718)
	γ	0.0987	0.1008	0.1000	(0.0715, 0.1268)
	ρ	0.2440	0.2473	0.2464	(0.1786, 0.3155)

Table 2: Posterior summaries of parameters from basic rejection ABC for $\varepsilon \in \{0.1, 0.01, 0.001\}$

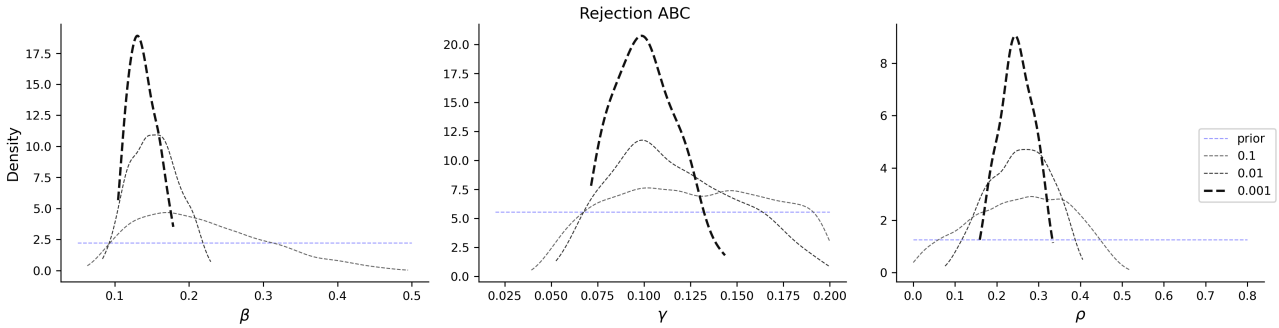

Figure 1: Posterior marginal distributions of parameters from rejection ABC with KDE.

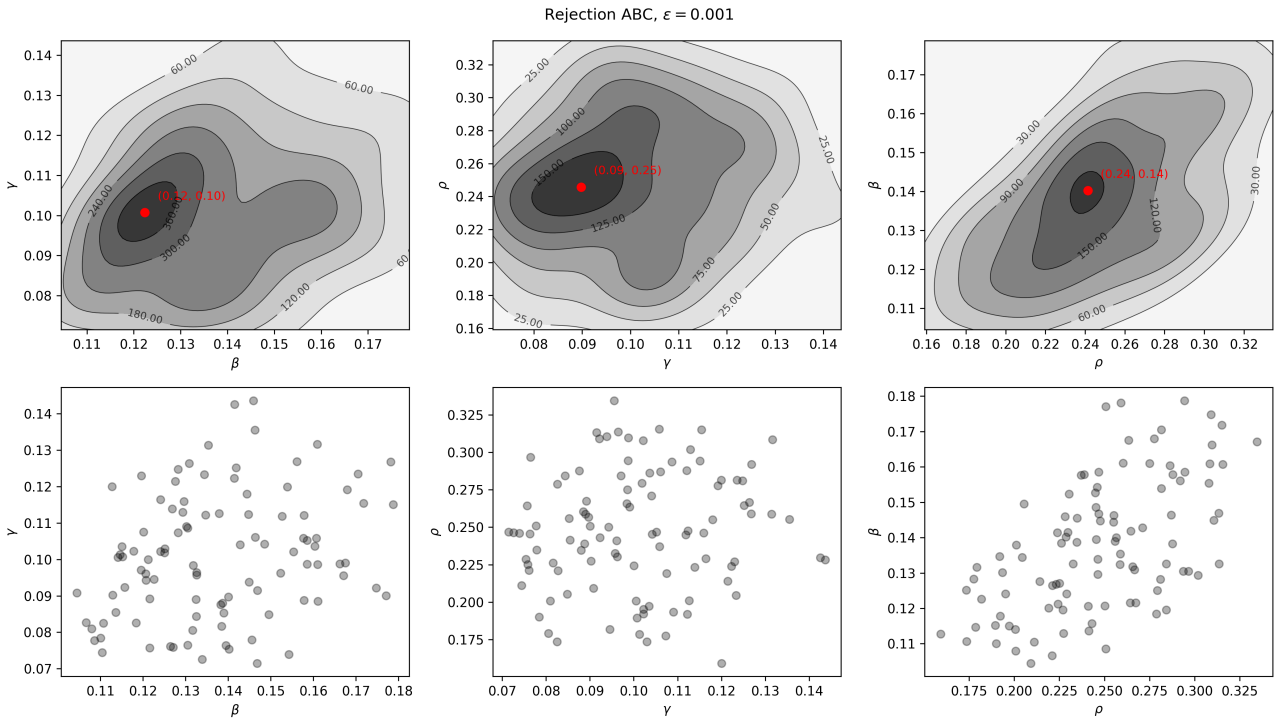

Figure 2: Joint posterior distributions of parameters from rejection ABC for $\varepsilon = 0.001$

§2.3 Validation

Posterior predictive checks were performed by simulating data from the accepted parameters and comparing the resulting summary statistics to those of the observed data (see The posterior predictive distributions of the summary statistics capture the observed values well, but the variability across simulations is large.

§3 Summary Statistics Design

The performance of ABC inference depends critically on the choice of summary statistics. In our setting, the relationship between the parameters and the observable data involves feedback between the epidemic dynamics and the network rewiring, making it difficult to identify informative summary statistic sets a priori. Using too many summary statistics can also degrade performance due to the curse of dimensionality, whereas using too few may lead to loss of information and biased inference.

To address this, we adopt a data-driven approach and consider two methods from the literature: (1) Partial least squares (PLS) regression and (2) regression-based summary construction. PLS constructs linear combinations of the candidate summaries that are maximally decorrelated and are also are highly correlated with the parameters (Wegmann et al., 2009). A reduction is achieved by selecting the top r components, where r is chosen via cross-validation, minimizing the root mean squared error of prediction (RMSEP). For regression-based summary construction, we follow the semi-automatic ABC framework of (Fearnhead & Prangle, 2012), who establish that the theoretically optimal summary statistics are the posterior means of the parameters. While these cannot be computed analytically, they can be estimated from simulated (θ, s) pairs, and the resulting estimates used as summary statistics within ABC. We approximate the posterior mean function using a random forest (Raynal et al., 2019) and a gradient boosting model, both fitted to the training simulations; the out-of-sample predicted parameter values then serve as the transformed summary statistics passed to the rejection and regression adjustment steps.

§3.1 Simulation Study

To assess the performance of these methods and the effect of the acceptance threshold ε, we conduct a simulation study. For each $\varepsilon \in \{0.001, 0.01, 0.1\}$, we generate 100 datasets with parameter values sampled from the prior distributionand perform ABC inference for each dataset.

§3.1.1 Methods

1. **Candidate Summary Statistics.** We construct a set of candidate statistics designed to capture both epidemic dynamics and adaptive network behaviour. For each simulation, we compute across the replicates the mean and standard deviation of the following features: peak infected fraction, time to peak infection, area under the infection curve, initial growth rate, total rewire count, peak rewire count, time to peak rewire count, the ratio of rewiring to infection, the lag between the infection and rewiring peaks, and the mean and standard deviation of the final degree distribution. This amounts to a total of 22 candidate summary statistics.
2. **Implementation.** All simulations and inference procedures use the prior described in Table 1. For the ABC rejection algorithm, we performed $N = 10^5$ simulations and stored these in a reference table. In the summary statistic selection step, we used $n = 10^4$ simulations drawn without replacement from the reference table. Post-rejection adjustment was performed by default via weighted local-linear regression with weights from an Epanechnikov kernel, following (Beaumont et al., 2002), without additional scaling of the parameters.
   (a) For the PLS method, we used the `PLSRegression` model from the `scikit-learn` library (Pedregosa et al. 2011) and followed Wegmann et al. (2009). Instead of a Box-Cox transformation, we applied a Yeo–Johnson power transformation followed by standardisation to zero mean and unit variance to the summary statistics prior to fitting the PLS regression model. This is because the Yeo–Johnson transformation does not require strictly positive inputs. The number of components to retain was chosen by 10-fold cross-validation based on the root mean squared error of prediction over a grid of up to 22 components; this procedure selected $r = 5$ components for `pls`. At inference time, the fitted PLS projection was applied to transform the observed and accepted summary statistics prior to distance computation and regression adjustment.
   (b) For the random forest method, we fitted one `RandomForestRegressor` model per parameter, each consisting of 100 regression trees, to the training statistics and parameter values. The fitted ensemble was then used to predict parameter values from summary statistics, and the resulting predicted values served as the transformed summary statistics passed to the rejection and adjustment steps. This construction implicitly performs a nonlinear, parameter-specific dimension reduction by mapping the high-dimensional summary vector to a scalar prediction for each parameter.
   (c) For the gradient boosting method, we proceeded analogously, fitting one `GradientBoostingRegressor` model per parameter with 100 boosting stages. As with the random forest approach, the predicted parameter values from the fitted ensemble were used as transformed summaries.

Both ensemble methods were implemented using the `scikit-learn` library, with all remaining hyperparameters set to their defaults. Standardization to mean zero and unit variance was applied to the summaries after transformation by the fitted ensemble.

3. **Evaluation metrics.** To assess estimation accuracy and posterior calibration, we report four metrics across 100 simulations. The relative absolute error of the mean integrated squared error (RAEMISE) measures the overall accuracy of the estimated posterior density relative to a baseline. The relative absolute error (RAE) of the posterior mode, mean, and median each measure point estimation accuracy for the corresponding summary of the posterior. Finally, posterior calibration is assessed via the Kolmogorov–Smirnov (KS) statistic, alongside the associated coverage p-value; a uniform distribution of posterior probabilities of the true parameter value is expected under a well-calibrated posterior, and significant deviations from uniformity indicate systematic bias or miscoverage.

Method	ε	Parameter	RARMISE[a]	RAE mean	RAE[b] mode	RAE median	KS_{100}[c]	Cov. P
all.	0.1	β	0.063 (0.039)	0.036 (0.036)	0.034 (0.030)	0.035 (0.035)	0.137	0.042*
		γ	0.125 (0.117)	0.049 (0.056)	0.043 (0.054)	0.045 (0.054)	0.203	0.000*
		ρ	0.156 (0.967)	0.077 (0.461)	0.078 (0.468)	0.078 (0.471)	0.109	0.174
	0.01	β	0.046 (0.026)	0.028 (0.026)	0.028 (0.026)	0.027 (0.025)	0.263	0.000*
		γ	0.074 (0.074)	0.035 (0.035)	0.035 (0.035)	0.033 (0.033)	0.209	0.000*
		ρ	0.032 (0.040)	0.015 (0.019)	0.015 (0.019)	0.015 (0.018)	0.168	0.006*
	0.001	β	0.039 (0.026)	0.025 (0.025)	0.025 (0.025)	0.025 (0.025)	0.233	0.000*
		γ	0.060 (0.061)	0.037 (0.048)	0.037 (0.048)	0.036 (0.049)	0.234	0.000*
		ρ	0.019 (0.014)	0.012 (0.010)	0.012 (0.010)	0.011 (0.010)	0.175	0.004*
pls	0.1	β	0.111 (0.070)	0.065 (0.059)	0.056 (0.049)	0.061 (0.056)	0.189	0.001*
		γ	0.151 (0.128)	0.071 (0.067)	0.068 (0.070)	0.068 (0.065)	0.207	0.000*
		ρ	1.019 (8.411)	0.805 (7.399)	1.096 (10.000)	0.939 (8.634)	0.076	0.579
	0.01	β	0.083 (0.045)	0.052 (0.045)	0.052 (0.045)	0.052 (0.045)	0.147	0.025*
		γ	0.099 (0.083)	0.058 (0.065)	0.058 (0.065)	0.057 (0.060)	0.185	0.002*
		ρ	0.093 (0.125)	0.051 (0.064)	0.051 (0.064)	0.045 (0.053)	0.094	0.326
	0.001	β	0.079 (0.047)	0.047 (0.048)	0.047 (0.048)	0.047 (0.047)	0.075	0.608
		γ	0.097 (0.094)	0.064 (0.083)	0.064 (0.083)	0.065 (0.083)	0.082	0.492
		ρ	0.062 (0.046)	0.037 (0.043)	0.037 (0.043)	0.038 (0.044)	0.055	0.910
rf	0.1	β	0.068 (0.040)	0.044 (0.040)	0.045 (0.041)	0.044 (0.039)	0.102	0.233
		γ	0.050 (0.074)	0.029 (0.071)	0.029 (0.072)	0.029 (0.071)	0.084	0.463
		ρ	0.158 (1.119)	0.049 (0.291)	0.047 (0.253)	0.041 (0.207)	0.109	0.175
	0.01	β	0.059 (0.026)	0.034 (0.029)	0.034 (0.029)	0.034 (0.029)	0.066	0.760
		γ	0.035 (0.045)	0.019 (0.032)	0.019 (0.031)	0.019 (0.032)	0.067	0.741
		ρ	0.034 (0.030)	0.019 (0.019)	0.019 (0.018)	0.019 (0.019)	0.083	0.469
	0.001	β	0.059 (0.028)	0.033 (0.031)	0.033 (0.031)	0.033 (0.031)	0.076	0.576
		γ	0.037 (0.061)	0.023 (0.051)	0.023 (0.051)	0.023 (0.055)	0.074	0.621
		ρ	0.027 (0.020)	0.016 (0.018)	0.016 (0.018)	0.017 (0.018)	0.064	0.784
gb	0.1	β	0.070 (0.041)	0.042 (0.040)	0.047 (0.039)	0.042 (0.039)	0.064	0.784
		γ	0.075 (0.115)	0.046 (0.109)	0.046 (0.108)	0.046 (0.108)	0.058	0.864
		ρ	0.501 (4.138)	0.431 (3.847)	0.426 (3.811)	0.424 (3.773)	0.078	0.547
	0.01	β	0.063 (0.029)	0.036 (0.031)	0.036 (0.031)	0.036 (0.031)	0.068	0.722
		γ	0.056 (0.057)	0.027 (0.033)	0.027 (0.033)	0.028 (0.034)	0.055	0.907
		ρ	0.062 (0.065)	0.032 (0.035)	0.032 (0.035)	0.031 (0.033)	0.061	0.832
	0.001	β	0.062 (0.031)	0.036 (0.033)	0.036 (0.033)	0.036 (0.033)	0.087	0.419
		γ	0.059 (0.075)	0.034 (0.056)	0.034 (0.056)	0.033 (0.051)	0.097	0.281
		ρ	0.046 (0.025)	0.025 (0.021)	0.025 (0.021)	0.026 (0.022)	0.106	0.201

RARMISE and RAE are given as the mean across 100 independent estimations with true values drawn from the prior (standard deviation in parentheses). *$P < 0.05$ without correction for multiple testing; cf. (see histograms).
[a] Relative absolute root mean integrated squared error (see text) with respect to the true value.
[b] Relative absolute error with respect to the true value.
[c] Kolmogorov–Smirnov distance between empirical distribution of posterior probabilities of the true parameter and U(0, 1).

Table 3: Accuracy of different methods for choosing summary statistics on a global scale.

§3.1.2 Results

We have compared the performance of ABC with summary statistics chosen via PLS, random forest (`rf`) and gradient boosting (`gb`) to that of ABC using all candidate summary statistics Table 3. The RAE behaved similarly for the three point estimates (mean, mode, and median) in good and bad regimes. For assessment, we sought low RARMISE and low RAE values, and require that the distribution of posterior probabilities of the true value does not deviate from uniform for any parameter. Across the compared methods, clear differences in both estimation accuracy and posterior calibration are observed. The `all` and `pls` approaches exhibit poor posterior calibration across $\varepsilon \in \{0.1, 0.01\}$, as reflected in low KS p-values and deviations from uniform coverage, indicating systematic bias and/or over- or under-confidence in the resulting posterior distributions. In contrast, both `rf` and `gb` yield substantially improved performance, but `rf` achieves lower RAE and RARMISE values across parameters, indicating more accurate and stable point and distributional estimates. Furthermore, `rf` shows consistently high KS p-values, suggesting well-calibrated posteriors that are consistent with the true parameter distributions. Overall, these results indicate that `rf` provides the most reliable trade-off between accuracy and calibration, and is therefore selected as the preferred method for subsequent analyses at the $\varepsilon = 0.01$ regime.

§3.1.3 Limitations

While the random forest and gradient boosting methods performed well in this simulation study, their performance may depend on the specific choice of hyperparameters (e.g., number of trees, depth of trees, learning rate) and the size of the training dataset used for fitting the models. More rigorous hyperparameter tuning could be conducted to potentially further improve performance, though this would come at the cost of increased compute. The regime with $\varepsilon = 0.001$ results in 100 accepted simulations, which may be insufficient for accurate density estimation, especially with a calibration test of size 100. However, the results for this regime are broadly consistent with those for $\varepsilon = 0.01$, which has a much larger number of accepted simulations, and the overall conclusions regarding the relative performance of the methods are unchanged.

In addition, it has been shown by (Aeschbacher et al., 2012) that locality refinements can improve point estimate accuracy in regression-based ABC settings. Aeschbacher et al. (2012) proposed an ABC framework with regression-based summary statistics and locality refinement, where the training samples are taken from an ε-neighbourhood around the observed summary statistics. This can improve the fidelity of posterior mean estimates in many settings. Nevertheless, even without this additional step, the resulting posterior approximations were well calibrated, with accurate parameter recovery across simulation scenarios.

§3.2 Application to Data

Posteriors inferred for the provided observations dataset with the explored methods and $\varepsilon = 0.01$ are shown in (Figure 3). The marginal posteriors are well within the prior support and are very concentrated. Point estimates and 95% highest posterior density (HPD) intervals obtained with `rf` and `gb` are reported in Table 4.

Method	Parameter	Mode	Mean	Median	95% HPD interval
`rf`	β	0.1416	0.1411	0.1413	(0.1287, 0.1563)
	γ	0.0876	0.0876	0.0876	(0.0856, 0.0896)
	ρ	0.2402	0.2404	0.2404	(0.2277, 0.2520)
`gb`	β	0.1460	0.1448	0.1452	(0.1287, 0.1606)
	γ	0.0857	0.0863	0.0863	(0.0829, 0.0897)
	ρ	0.2478	0.2487	0.2482	(0.2297, 0.2691)

Table 4: Posterior estimates for the provided data from ABC with summary statistics chosen globally via random forests (`rf`) and gradient boosting (`gb`) with acceptance rate $\varepsilon = 0.01$.

The marginal posterior obtained by `rf` is more concentrated than that obtained by `gb` and there are some differences in the point estimates and HPD intervals. Compared to 'gb', `rf` thinks that β is slightly lower, γ is slightly higher, and ρ is slightly lower.

§3.2.1 Feature Importance

To investigate which summary statistics were most influential in the random forest-based inference, we compare the feature importance scores from the fitted random forest models given in Table 5. The most influential statistic for β was `mean:initial_growth_rate` with a very high score of 0.990, capturing the infection rate signal without being confounded by the rewiring dynamics. For γ, the most influential statistic was `mean:auc_infected_fraction` (0.782) followed by `sd:auc_infected_fraction` (0.119), and for ρ,

`mean:sd_degree` (0.735) followed by `mean:peak_rewire_count` (0.159). Other summaries are not necessarily uninformative or insufficient but feature importance scores are relative and depend on the fitted ensemble. Therefore, it should be interpreted as a model-dependent measure rather than a structural statement about sufficiency.

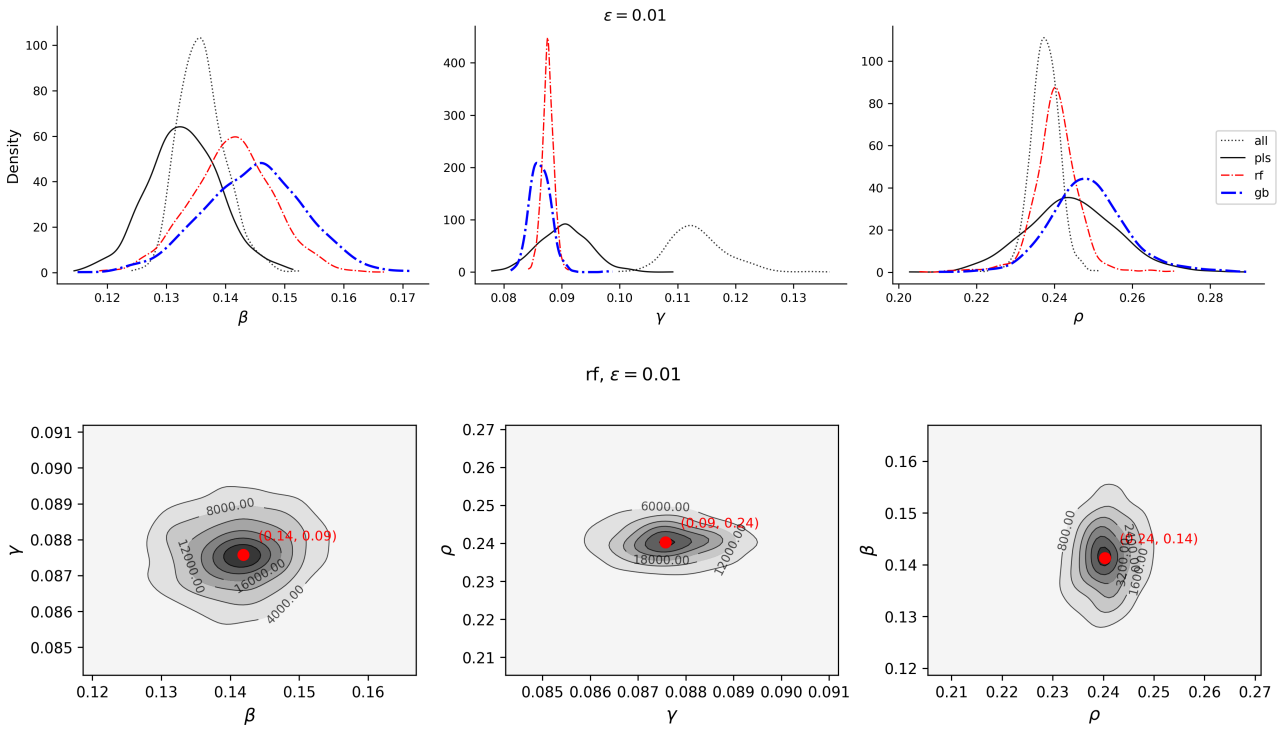

Figure 3: Marginal and joint pairwise posterior distributions inferred from the provided data. Posteriors obtained with tolerance $\varepsilon = 0.01$ and various methods for choosing summary statistics are compared. The dot-dashed red line corresponds to the method that performed best in the simulation study (`rf`; Table 3). Point estimates and 95% HPD intervals are given in Table 4

Summary statistic	β	γ	ρ
`mean:auc_infected_fraction`	0.000971	0.782088	0.000713
`sd:auc_infected_fraction`	0.000784	0.119448	0.000257
`mean:peak_infected_fraction`	0.000415	0.008917	0.000790
`sd:peak_infected_fraction`	0.000449	0.009251	0.000251
`mean:time_to_peak_infected_fraction`	0.000380	0.000675	0.010183
`sd:time_to_peak_infected_fraction`	0.000304	0.000443	0.000380
`mean:initial_growth_ratio`	0.990066	0.001505	0.025429
`sd:initial_growth_ratio`	0.000656	0.003094	0.000350
`mean:peak_rewire_count`	0.000295	0.000369	0.159044
`sd:peak_rewire_count`	0.000289	0.000530	0.009788
`mean:time_to_peak_rewire_count`	0.000192	0.039617	0.024164
`sd:time_to_peak_rewire_count`	0.000334	0.000528	0.000433
`mean:total_rewire_count`	0.000148	0.006622	0.001383
`sd:total_rewire_count`	0.000185	0.006886	0.000797
`mean:mean_degree`	0.000718	0.000432	0.000147
`sd:mean_degree`	0.000586	0.000442	0.000135
`mean:sd_degree`	0.000255	0.000791	0.735150
`sd:sd_degree`	0.000447	0.003894	0.000846
`mean:rewire_to_infection_ratio`	0.000200	0.009692	0.002091
`sd:rewire_to_infection_ratio`	0.000196	0.000928	0.000506
`mean:lag_peak`	0.001677	0.003284	0.026858
`sd:lag_peak`	0.000453	0.000564	0.000304

Table 5: Feature importance scores from the fitted random forest models used for summary statistic construction. The most influential statistics for each parameter are highlighted.

§4 Advanced Methods

§4.1 Neural Posterior Estimation

While the semi-automatic ABC framework constructs summary statistics via regression-based dimension reduction, the rejection step remains computationally wasteful: the vast majority of simulations are discarded, and inference must be repeated from scratch for each new observation. Neural Posterior Estimation (NPE) addresses both limitations by training a neural density estimator to directly approximate $p(\theta|s)$ from simulated (θ, s) pairs, producing an amortised posterior that can be evaluated at any observation without retraining.

NPE parametrises the posterior using a normalising flow, a composition of invertible, differentiable transformations that maps a simple base distribution (typically Gaussian) to a flexible target density. Because the transformations are invertible, both sampling and exact density evaluation are tractable, and the flow parameters are learned by maximum likelihood on simulated (θ, s) pairs. This allows the model to represent complex, multimodal, and correlated posteriors that are out of reach for Gaussian or rejection-based approximations.

§4.1.1 Setup

We model the posterior using a Masked Autoregressive Flow (MAF) (Papamakarios et al., 2017), in which each transformation is an autoregressive network that conditions each output dimension on all previous ones, enabling expressive density estimation while preserving tractable likelihoods. The flow is fitted via Automatic Posterior Transformation (APT) (Greenberg et al., 2019), as implemented in the `sbi` library (Tejero-Cantero et al., 2020). APT trains the flow by maximising the probability of simulated parameters under a dynamically updated proposal, avoiding the importance weights and post-hoc corrections that destabilise earlier sequential NPE approaches.

Training used $N = 10^4$ parameter–summary pairs drawn from the reference table, with the full summary vector of size 22 as input. No hyperparameter tuning was performed, and the default MAF architecture of 5 autoregressive layers with 50 hidden units each was used throughout, converging after 211 epochs (~ 2 minutes).

APT-MAF has been demonstrated on inputs of dimension 10^4 without any summary statistic construction (Greenberg et al., 2019), so its possible to use the raw time series and final degree histogram as inputs. This may allow the model to learn more informative features than the hand-crafted summaries, but will incur greater computational cost. As we will see, the current setup already achieves good performance with a fraction of the simulation budget required by ABC.

§4.1.2 Posterior inference

The posterior was evaluated by drawing $n = 10^3$ samples from the fitted density estimator conditioned on s_{obs}, requiring no rejection step or tolerance threshold. Marginal posteriors for all methods except pls are shown in Figure 4. APT-MAF closely follows the `rf` in the marginal posterior shape for γ, but the mode sits slightly higher. Compared to `rf`, APT-MAF gives lower estimates of β and higher estimates of ρ, with more concentrated marginals.

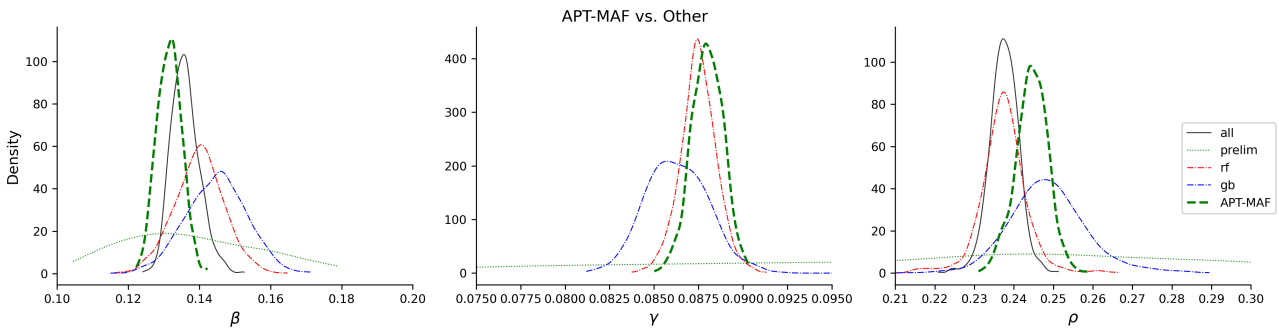

Figure 4: Marginal posteriors for β, γ, and ρ under basic rejection ABC, `all`, `rf`, `gb` and `APT-MAF`.

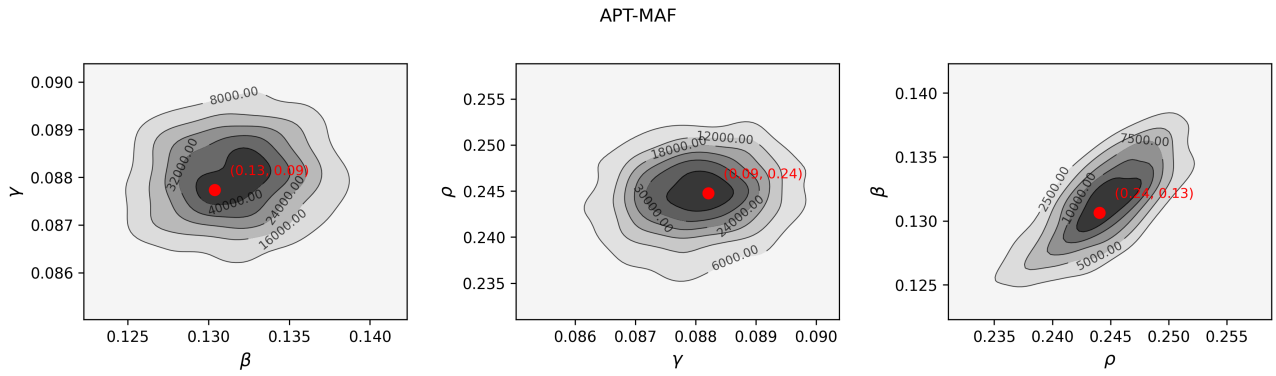

Figure 5: Pairwise joint posterior from APT-MAF, constructed from $n = 1000$ samples.

The number of simulation pairs used to train the NPE is much less than basic rejection ABC and its variants e.g. ABC-SMC. These methods may have smaller per-round compute budgets, but may still require large total simulation counts to reach small tolerances (Sisson et al., 2007).

§4.1.3 Calibration

Calibration was assessed via simulation-based calibration (SBC) (Talts et al., 2018) on 100 held-out parameter–summary pairs, drawing $n = 10^3$ posterior samples per observation. We report the KS statistic against uniformity of ranks, the Classifier Two-Sample Test (C2ST), a binary classifier accuracy near 0.5 indicates the posterior and reference samples are indistinguishable and the Data-Averaged Posterior (DAP) statistic as a complementary coverage measure.

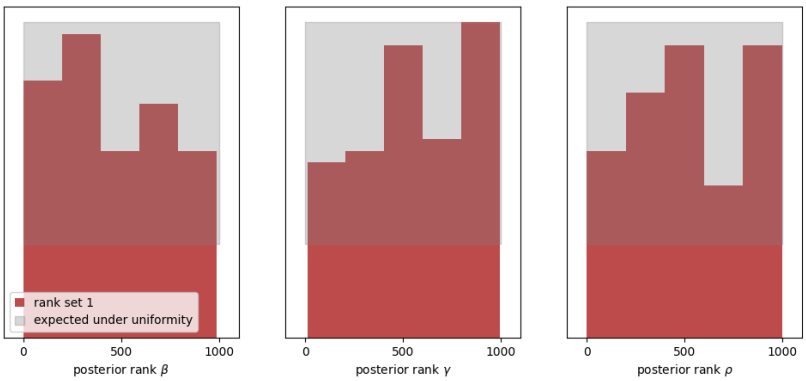

Figure 6: SBC rank histograms for β, γ, ρ. Dashed line indicates perfect calibration.

The rank histograms (Figure 6) are broadly uniform across all three parameters, and all KS p-values sit comfortable above the significance level. The C2ST statistics deviate little from 0.5, indicating no detectable systematic bias in the learned posterior. This compares favourably to rejection ABC, where coverage p-values were near zero for most parameter–method combinations, and is broadly competitive with `rf`. However, the DAP statistic is somewhat on the low side for all three parameters. This could be an artifact of the relatively small number of posterior samples and calibration observations.

Parameter	Mode	Mean	Median	95% HPD interval	KS_{100} Cov. P	C2ST ranks	C2ST DAP
β	0.1322	0.1315	0.1316	(0.1254, 0.1385)	0.2527	0.42	0.39
γ	0.0879	0.0880	0.0880	(0.0862, 0.0896)	0.1577	0.545	0.37
ρ	0.2443	0.2445	0.2446	(0.2362, 0.2524)	0.7511	0.51	0.33

Table 6: APT-MAF posterior estimates and SBC calibration statistics.

Overall, APT-MAF achieves well-calibrated, concentrated posteriors for all three parameters using a fraction of the simulation budget required by rejection-based methods, and without explicit summary statistic construction or tolerance selection. The results suggest that neural posterior estimation is a viable and efficient alternative to ABC for this class of adaptive-network models. Compared to semi-automatic ABC, NPE methods are easily scalable to high-dimensional inputs such as time series.

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§A Appendix

The code used for all simulations and analyses in this report is available at my github repository along with documentation for reproducing the results.

Name	Formula	Description	Intuition
Peak infected fraction	$\max_t I_t$	Max infected fraction	Outbreak severity (β, ρ)
Time to peak infection	$\arg\max I_t$	Time of peak infection	Epidemic speed indicator
Area under infection curve	$\sum_t I_t$	Integral of infected fraction over time	Total epidemic burden
Initial growth ratio	I_1/I_0		Early transmission rate (β-driven)
Total rewire count	$\sum_t R_t$	Total rewiring events	Overall behavioral adaptation (ρ)
Peak rewire count	$\max_t R_t$	Max rewiring in a single step	Peak avoidance intensity
Time to peak rewire	$\arg\max R_t$	Time of peak rewiring	Timing of behavioral response
Rewire / infection ratio		Total rewiring / total infections	Adaptation vs transmission strength
Peak lag	$\max_t I_t - \max_t R_t$	Time difference between peaks	Response delay of behavior
Mean final degree	μ_{deg}	Mean node degree at final time	Residual connectivity
SD final degree	σ_{deg}	Std. dev. of final degree	Network heterogeneity (ρ-driven)

Table 7: Summary statistics, their descriptions, and intuition.